

PART-A

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INSTRUCTIONS

1. Check that you have 10 printed pages. The PART-A of the examination has 3 questions, for a total of 20 points.
2. Attempt all questions.
3. There is no credit for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the space marked for answers. Any answer written in any space other than the designated space will not be evaluated.
5. The pages numbered 8 to 10 are kept as additional pages. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	0	9	14

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$Y_1 = \theta_1 + \theta_2 + \epsilon_1,$$

$$Y_2 = \theta_1 - 2\theta_2 + \epsilon_2,$$

$$Y_3 = 2\theta_1 - \theta_2 + \epsilon_3,$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

$$Y = X\beta + \epsilon$$

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \end{bmatrix}$$

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

$$X^T X = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}, (X^T X)^{-1} = \frac{1}{36-9} \begin{bmatrix} 6 & +3 \\ +3 & 6 \end{bmatrix}$$

$$= \frac{1}{9} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$(X^T X)^{-1} X^T = \frac{1}{9} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 3 & 0 & 3 \\ 3 & -3 & 0 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 1/3 & 0 & 1/3 \\ 1/3 & -1/3 & 0 \end{bmatrix}$$

$$\hat{\beta} = (X^T X)^{-1} X^T Y = \begin{bmatrix} 1/3 & 0 & 1/3 \\ 1/3 & -1/3 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1/3 y_1 + 1/3 y_3 \\ 1/3 y_1 - 1/3 y_2 \end{bmatrix}$$

So $\hat{\beta}_1$ or $\hat{\theta}_1 = \underline{\underline{1/3 y_1 + 1/3 y_3}}$ and $\hat{\theta}_2 = \underline{\underline{1/3 y_1 - 1/3 y_2}}$

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Ex: interval

$$[X, Y, Z] = 0$$

$$\begin{array}{c|c|c} X & Y & Z \\ \hline 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{array}$$

$$[X, Y, Z] = 0 \quad [X, Y, Z] = 0$$

Ex: interval

$$[X, Y, Z] = 0 \quad [X, Y, Z] = 0 \quad [X, Y, Z] = 0 \quad [X, Y, Z] = 0$$

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Ex: interval

$$[X, Y, Z] = 0 \quad [X, Y, Z] = 0 \quad [X, Y, Z] = 0 \quad [X, Y, Z] = 0$$

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Ex: interval

$$[X, Y, Z] = 0 \quad [X, Y, Z] = 0 \quad [X, Y, Z] = 0 \quad [X, Y, Z] = 0$$

2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\mathbf{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.

(a) (3 points) Compute the correlation matrix ρ of $\mathbf{X}$.

(b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

$$\mathbf{X} = (X_1, X_2, X_3)'$$

$$\begin{array}{ccc} X_1, X_2, X_3 & & U_1, U_2, \dots \sim U(0,1) \\ \downarrow \quad \downarrow \quad \downarrow & & \\ U_1 & U_2 & U_1 + U_2 \end{array}$$

Covariance matrix Σ of $\mathbf{X}$.

Cov Matrix Σ

	X_1	X_2	X_3
X_1	$1/2$	$1/2$	$1/6$
X_2	$1/2$	$1/2$	$1/6$
X_3	$1/6$	$1/6$	$1/3$

$$\text{Var}(X_1) = E[X_1^2] - (E[X_1])^2 = \int_0^1 x^2 dx - \left(\int_0^1 x dx\right)^2 = \left[\frac{x^3}{3}\right]_0^1 - \left(\left[\frac{x^2}{2}\right]_0^1\right)^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

$$\text{Var}(X_2) = \text{Var}(X_1) = 1/12$$

$$\begin{aligned} \text{Var}(X_3) &= E[X_3^2] - (E[X_3])^2 = E[(U_1 + U_2)^2] - (E[U_1 + U_2])^2 = E[U_1^2 + U_2^2 + 2U_1U_2] \\ &\quad - (E[U_1 + U_2])^2 \\ &= \frac{1}{3} + \frac{1}{3} + 2 \cdot \frac{1}{3} - \left(\frac{1}{2} + \frac{1}{2}\right)^2 = \frac{4}{3} - 1 = \frac{1}{3} \end{aligned}$$

$$\begin{aligned} \text{Cov}(X_1, X_2) &= E[X_1 X_2] - E[X_1]E[X_2] \\ &= \frac{1}{3} - \frac{1}{2} \times \frac{1}{2} = \frac{1}{3} - \frac{1}{4} = \frac{1}{12} \end{aligned}$$

$$\begin{aligned} \text{Cov}(X_1, X_3) &= \cancel{E[X_1 X_3]} = E[X_1 X_3] - E[X_1]E[X_3] = \frac{1}{3} + \frac{1}{3} - \frac{1}{2} = \frac{2}{3} - \frac{1}{2} = \frac{1}{6} \\ &\quad \downarrow \\ &\quad E[U_1(U_1 + U_2)] \end{aligned}$$

$$\text{Cov}(X_2, X_3) = \text{Cov}(X_1, X_3) = 1/6$$

Correlation matrix ρ of $\mathbf{X}$

$$\text{Corr}(X_1, X_1) = \text{Corr}(X_2, X_2) = \text{Corr}(X_3, X_3) = 1$$

$$\text{Corr}(X_1, X_2) = \frac{\text{Cov}(X_1, X_2)}{\sqrt{\text{Var}(X_1)}\sqrt{\text{Var}(X_2)}} = \frac{1/12}{\sqrt{1/12}\sqrt{1/12}} = \frac{1/12}{1/12} = 1$$

$$\text{Corr}(X_1, X_3) = \frac{\text{Cov}(X_1, X_3)}{\sqrt{\text{Var}(X_1)}\sqrt{\text{Var}(X_3)}} = \frac{1/6}{\sqrt{1/12}\sqrt{1/3}} = \frac{1/6}{1/6} = 1$$

$$\text{Corr}(X_2, X_3) = \frac{\text{Cov}(X_2, X_3)}{\sqrt{\text{Var}(X_2)}\sqrt{\text{Var}(X_3)}} = \frac{1/6}{\sqrt{1/2} \times \sqrt{1/3}} = \frac{1/6}{1/6} = 1$$

So correlation matrix of X

	X_1	X_2	X_3
X_1	1	1	1
X_2	1	1	1
X_3	1	1	1

b) Finding first principal component:- $\det(\beta - \lambda I) = 0$

$$\det \begin{vmatrix} 1-\lambda & 1 & 1 \\ 1 & 1-\lambda & 1 \\ 1 & 1 & 1-\lambda \end{vmatrix} = 0 \rightarrow (1-\lambda)[(1-\lambda)^2 - 1] - 1[1-\lambda-1] + 1[1-1+\lambda]$$

$$= (1-\lambda)(1-\lambda)^2 - 1 + \lambda + \lambda + \lambda = 1 - \lambda^3 - 3\lambda + 3\lambda^2 - 1 + 3\lambda$$

$$= 3\lambda^2 - \lambda^3 = 0 \rightarrow \lambda^2(3-\lambda) = 0$$

$$\lambda_1 = 3, \lambda_2 = 0, \lambda_3 = 0$$

Variance of first principle component as a ratio of total variance

$$= \frac{\lambda_1}{\sum \lambda_i} = \frac{3}{3+0+0} = 1$$

eigen vector corresponding to 1st PCA :- $\left[\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right]$

$$Y_1 = X_1'X$$

Total variance = $\sum \text{Var}(X_i)$

$$\text{Trace}(\Sigma) = \sum \text{Var}(Y_i)$$

So variance of first principle component $\Rightarrow$

$$= 1 \times \text{Trace} = \frac{1}{2}$$

3. Let the matrix of distance be

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

(a) (6 points) Cluster the four items using each of the following procedures:

- Complete linkage hierarchical procedure.
- Average linkage hierarchical procedure.

(b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

a) i) Complete linkage hierarchical procedure :- The distance between the clusters is taken as the maximum i.e.

$$\text{dist}(uv), w = \max \{ \text{dist}_{u,w}, \text{dist}_{v,w} \}$$

Initial distance matrix. (smallest distance → most similar)

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

smallest distance is $d_{12}=1$
so 1 & 2 form a cluster

	(1,2)	3	4
(1,2)	0		
3	11	0	
4	5	3.5	0

smallest distance is $d_{34}=3.5$
so 3 & 4 form a cluster.

	(1,2)	(3,4)
(1,2)	0	
(3,4)	11	0

$\max \{ d_{(1,2),3} \text{ \& } d_{(1,2),4} \}$

Hence the clusters formed are (1,2) and (3,4) and distance b/w them = 11.

ii) Average linkage hierarchical procedure :- The distance b/w clusters is taken as the average of all distances b/w elements of the clusters.

$$\text{dist}_{u,v} = \frac{1}{|U||V|} \sum_{i \in U} \sum_{j \in V} d_{ij}$$

Initial distance matrix

$$\begin{array}{c}
 \begin{matrix} & 1 & 2 & 3 & 4 \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 11 & 5 \\ & 0 & 2 & 3 \\ & & 0 & 2.5 \\ & & & 0 \end{bmatrix} \end{matrix}
 \end{array}$$

Smallest distance = $d_{1,2} = 1$
So 1 & 2 form a cluster.

$$\begin{array}{c}
 \begin{matrix} & (1,2) & 3 & 4 \\ \begin{matrix} (1,2) \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & & \\ 6.5 & 0 & \\ 4 & 2.5 & 0 \end{bmatrix} \end{matrix}
 \end{array}$$

$$\begin{array}{c}
 \begin{matrix} & (1,2) & (3,4) \\ \begin{matrix} (1,2) \\ (3,4) \end{matrix} & \begin{bmatrix} 0 & \\ 5.25 & 0 \end{bmatrix} \end{matrix}
 \end{array}$$

distances updation
Smallest distance is $d_{3,4}$ so 3, 4 will form a cluster.

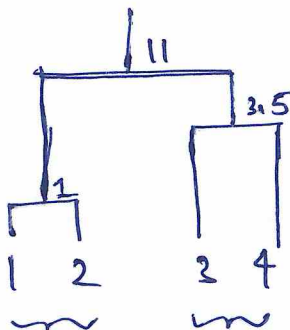
$$\begin{aligned}
 d_{(1,2),3} &= \frac{1}{2} [d_{1,3} + d_{2,3}] \\
 &= \frac{1}{2} [11 + 2] = 6.5
 \end{aligned}$$

$$\begin{aligned}
 d_{(1,2),4} &= \frac{1}{2} [d_{1,4} + d_{2,4}] \\
 &= \frac{1}{2} [5 + 3] = 4
 \end{aligned}$$

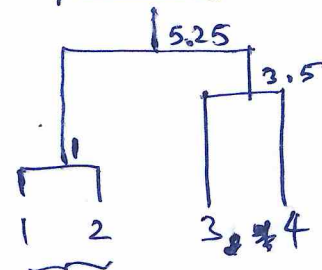
$$\begin{aligned}
 d_{(3,4),(1,2)} &= \frac{1}{4} (d_{1,3} + d_{1,4} + d_{2,3} + d_{2,4}) \\
 &= \frac{1}{4} (11 + 5 + 2 + 3) \\
 &= 5.25
 \end{aligned}$$

Hence the clusters formed are
(3, 4) and (1, 2) & distance b/w them = 5.25

b) Dendograms
for complete linkage hierarchical procedure



For average linkage hierarchical procedure



The results of clustering are similar, both the procedures result in the formation of same clusters, though the distance b/w them is different.

ADDITIONAL PAGES FOR ANSWERS

